

Comprehensive Test Coverage Audit Report

Driftless (FocusBuddy) | March 19, 2026

Principal QA Auditor | Static Analysis Only | No Tests Executed

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1. Executive Summary

Metric	Value
Total features identified	57 distinct functional components
Test suites	30 Jest suites, 461 tests, all passing
Deno test suites	3 suites, 48 tests (require deno runtime)
Total test count	509 individual tests across 33 files
Files with unit tests	25 of 57 (~44%)
Statement coverage (src/)	74.40% (683/918)
Branch coverage (src/)	59.38% (348/586)
Edge Functions tested	2 of 6 (partial — pure function tests only, no HTTP handler tests)
Screen/component tests	3 of 28 components (10.7%), 0 of 26 screens
E2E / integration tests	0
CI coverage threshold	70% minimum statements (enforced)
Overall system risk	MEDIUM

Brutal Truth: The service and store layers are well-tested (93% and 87% respectively), and recent additions have closed many critical gaps (REFUND webhook handler, prompt injection defense, restoreTimerFromMMKV, signUp, route protection logic, timeout utility). However, the 4 untested Edge Functions (ai-checkin, activate-trial, data-export, delete-account) representing 907 lines of server-side logic remain the largest risk. All 26 app screens have zero rendering tests. The system is significantly stronger than before but still has meaningful server-side blind spots.

Layer	Files	Tests	Coverage	Risk
Zustand Stores (5)	5 of 5	138 tests (7 suites)	87%	LOW
Services (11)	10 of 11	~110 tests	93%	LOW
Libs/Utils (5)	4 of 5	~92 tests	~70%	LOW
Utils (2)	2 of 2	~76 tests	~99%	LOW
Components (17)	3 of 17	~36 tests	<10%	MEDIUM
Hooks (1)	1 of 1	7 tests	100%	LOW
Screens (26)	0 of 26	0 rendering tests	0%	HIGH
Edge Functions (6)	2 of 6 (partial)	48 pure-fn tests	~20%	HIGH
E2E/Integration	N/A	0 tests	0%	HIGH

2. Feature Coverage Table

Stores (State Management)

Feature	Location	Tested ?	Types	What Tested	Not Tested	Risk
Auth Store	stores/authStore	Yes	Unit	signIn, signOut, signUp, resetPassword, fetchProfile, updateProfile (19 tests)	initialize() callback behavior, onAuthStateChange lifecycle	Med
Chat Store	stores/chatStore	Yes	Unit	fetch, send, delete, clear, rateLimit (29 tests, 100% stmt)	Timeout behavior in context, concurrent sends	Low
Goal Store	stores/goalStore	Yes	Unit	fetch, create, archive, toggleTask (26 tests)	Unauthenticated paths (5), fetchTodayCheckIn/Tasks error	Low
Session Store	stores/sessionStore	Yes	Unit	start, pause, resume, end, updateElapsed, restoreTimerFromMMKV (26 tests)	All 9 restore branches now tested	Low
Subscription Store	stores/subscriptionStore	Yes	Unit	init, entitlement, offerings, purchase, restore, deriveFromCustomerInfo (40 tests)	Happy init path (mock limitation)	Low

Services (Data Layer)

Feature	Location	Tested ?	Types	What Tested	Not Tested	Risk
Auth Service	services/auth	Yes	Unit	signIn+profile, signIn, signOut, reset, getSession (9 tests)	onAuthStateChange listener	Low
Chat Service	services/chat	Yes	Unit	send, getMessages, delete, clear (11 tests)	None significant	Low
Check-in Service	services/checkin	Yes	Unit	getTodayCheckIn, startCheckIn+streak, getRecent (8 tests)	90-day cap enforcement	Low
Goals Service	services/goals	Yes	Unit	getActive, create+deactivate, archive, getGoals (7 tests)	Archived filter	Low
Profile Service	services/profile	Yes	Unit	get, update whitelist, completeOnboarding CAS, delete (10 tests)	signOut error in delete	Low
Promises Service	services/promises	Yes	Unit	CRUD, trustScore formula, history clamping (15 tests)	broken_reason now properly cleared when kept=true	Low
Sessions Service	services/sessions	Yes	Unit	create, update whitelist, end, recent, stats (11 tests)	Stats with 0 sessions	Low
Shields Service	services/shields	Yes	Unit	award CAS, use CAS, getCount (10 tests)	null vs empty-array edge	Low

Feature	Location	Tested ?	Types	What Tested	Not Tested	Risk
Subscription Svc	services/subscri ption	Yes	Unit	get, update safe fields, trialStatus (13 tests)	Explicit whitelist proof	Low
Tasks Service	services/tasks	Yes	Unit	getForCheckIn, getToday, toggle, create (11 tests)	None significant	Low

Edge Functions (Server-side, Deno Runtime)

Feature	Location	Tested ?	Types	What Tested	Not Tested	Risk
ai-chat	supabase/fn/ai- chat	Partial	Deno	sanitizeForPrompt (11), detectShameLanguage (8), timezone, rate limit constant	HTTP handler, OpenAI call, DB persist, auth, timeout	Med
ai-checkin	supabase/fn/ai- checkin	No	None	Nothing	ALL: auth, rate limit, input validation, OpenAI, response parsing, check-in upsert, task insert (444 lines, ~21 branches)	CRIT
activate-trial	supabase/fn/acti vate-trial	No	None	Nothing	ALL: auth, rate limit, CAS guard, already-subscribed 409, INSERT fallback (155 lines, ~10 branches)	High
revenuecat-webh ook	supabase/fn/rev enuecat-webho ok	Partial	Deno	signPayload (3), event mapping (10 types incl REFUND+PRODUCT_ CHANGE), UUID regex (3)	HTTP handler, HMAC verification, idempotency, DB upsert (317 lines)	Med
delete-account	supabase/fn/del ete-account	No	None	Nothing	ALL: auth, rate limit, soft-delete, sign-out (119 lines, ~8 branches)	Med
data-export	supabase/fn/dat a-export	No	None	Nothing	ALL: auth, rate limit, 8-table fetch, JSON response (192 lines, ~7 branches)	Med

Screens (UI + Business Logic)

Feature	Location	Tested ?	Types	What Tested	Not Tested	Risk
Root Layout	app/_layout	Partial	Logic	Route protection logic (10 tests), 30-day auto-logout (5 tests)	Rendering, PASSWORD_RECOVERY deep link, theme, SplashScreen	Med
Home Screen	app/(tabs)/index	No	None	Nothing	Task toggle, data fetching, goal display, refresh (346 lines)	Med
Progress Screen	app/(tabs)/progress	No	None	Nothing	Period selector, stats, session history, FlatList (812 lines)	Low
Chat Screen	app/(tabs)/chat	No	None	Nothing	Optimistic UI, rate limit display, load-more, prefill, shame redirect (348 lines)	Med
Settings Screen	app/(tabs)/settings	No	None	Nothing	Toggle switches, sign-out, navigation (439 lines)	Low
Check-In Screen	app/check-in	No	None	Nothing	Energy select, AI task gen, clarification loop (now capped at 3), task editing (861 lines)	High
Timer Screen	app/timer	Partial	Logic	formatStartTime (5), goal-reached (3), elapsed capping (2), cold-start (2), setup validation (4)	Rendering, tick interval, AppState reconciliation (374 lines)	Med
Session Rating	app/session-rating	Partial	Logic	Rating map (4), formatDuration (5), auth guard (2), save branching (2)	Rendering, app-quit fallback integration, streak update (294 lines)	Med
Promises Screen	app/promises	No	None	Nothing	Promise CRUD, trust score display, history (423 lines)	Med
Paywall Screen	app/paywall	Partial	Logic	deriveFromCustomerInfo (7), store state (5), result structures (5)	Rendering, purchase flow integration (175 lines)	Med
Shame Emergency	app/shame-emergency	No	None	Nothing	Evidence fetch, micro-actions, grounding flow (420 lines)	Low
Edit Profile	app/edit-profile	No	None	Nothing	Profile form, goal display (286 lines)	Low
Bad Day Toolbox	app/bad-day-toolbox	No	None	Nothing	Category display, action selection (225 lines)	Low
Data Export	app/data-export	No	None	Nothing	Export trigger, share flow (170 lines)	Low
Delete Account	app/delete-account	No	None	Nothing	Confirmation, soft-delete, sign-out (133 lines)	Med

Infrastructure

Feature	Location	Tested ?	Types	What Tested	Not Tested	Risk
MMKV storage	lib/mmkv	Yes	Unit	Timer state R/W, activity tracking, parse error (24 tests)	None significant	Low
Sentry	lib/sentry	Yes	Unit	init, captureError, setUser, breadcrumb (13 tests)	setSentryEnabled, beforeSend scrub	Low
Timeout utility	lib/timeout	Yes	Unit	withTimeout, TimeoutError, timer cleanup, PromiseLike (9 tests)	None significant	Low
fireAndForget	lib/fireAndForget	No	None	Nothing	Promise rejection capture to Sentry	Low
Supabase client	lib/supabase	No	None	Nothing	Client init, secure-store adapter, env validation	Low
useAppState hook	hooks/useAppState	Yes	Unit	Foreground, background, cleanup (7 tests)	None	Low
Time utils	utils/time	Yes	Unit	All formatters and helpers (~76 tests)	None	Low
Constants	utils/constants	Yes	Unit	All values/structure validated (~26 tests)	None	Low
CI/CD Pipeline	.github/workflows/ci.yml	N/A	N/A	5 jobs: tsc, lint, test, security, build + 70% coverage threshold	No Deno test job	Med
DB Migrations	supabase/migrations/	Partial	Deno	RLS on 8 tables, auth.uid() SELECT policies, atomic functions exist (8 tests)	UPDATE/DELETE policies, atomic function behavior	Med

3. Untested / Weakly Tested Features (Detailed)

ai-checkin Edge Function (444 lines)

Feature	Priority	Test Type	Missing Scenarios	Failure Impact
ai-checkin Edge Function	CRITICAL	Unit+Integration	Auth, rate limiting, energy level validation (1-5), message/hardReason length caps, parseAiResponse (JSON extraction from markdown fences + fallback), clarification vs task response branching, check-in upsert, task row batch insert, partial success handling, OpenAI timeout, 15s AbortController	Invalid AI response crashes check-in flow. Task generation with malformed JSON breaks daily routine. Uncapped input fields enable payload attacks. Clarification infinite-loop now capped at 3 rounds client-side, but server has no cap.

activate-trial Edge Function (155 lines)

Feature	Priority	Test Type	Missing Scenarios	Failure Impact
activate-trial Edge Function	HIGH	Unit	Auth, rate limiting (5/hr), atomic CAS guard (.not("status","in","(active,trialing)")), already-subscribed 409 response, missing subscription row INSERT fallback, trial end date calculation	Rate limit bypass enables trial farming. CAS guard regression allows duplicate active trials. INSERT fallback failure leaves user in limbo.

authStore.initialize() callback (lines 64-79)

Feature	Priority	Test Type	Missing Scenarios	Failure Impact
authStore.initialize() callback	HIGH	Unit	onAuthStateChange callback invocation: session present → fetchProfile fire-and-forget, session null → clear profile, captureError on profile fetch failure	Auth lifecycle hook tested only for return type, not actual behavior. Profile never loads if callback breaks. Silent failures in profile fetch leave stale data.

data-export Edge Function (192 lines)

Feature	Priority	Test Type	Missing Scenarios	Failure Impact
data-export Edge Function	MEDIUM	Unit	Auth, rate limit, 8-table fetch with row cap (10,000), JSON response structure, null data handling, Content-Disposition headers	Export returns malformed data if any table query fails. Row cap not enforced. GDPR compliance risk.

delete-account Edge Function (119 lines)

Feature	Priority	Test Type	Missing Scenarios	Failure Impact
delete-account Edge Function	MEDIUM	Unit	Auth, rate limiting (3/hr), soft-delete profile, sign-out after deletion, error recovery	Rate limit bypass enables deletion harassment. Soft-delete failure leaves account in inconsistent state.

Check-In Screen (861 lines, ~38 branches)

Feature	Priority	Test Type	Missing Scenarios	Failure Impact
Check-In Screen	MEDIUM	Component	Energy selection flow, AI task generation, clarification loop (now capped at 3), task editing/reordering, handleConfirm navigation, handleRestDay shield CAS, custom time input validation (5-180)	Clarification loop cap only tested via static code review, not runtime. Shield CAS failure path not rendered. Custom time bypasses 5-180 range validation display.

Database Migrations / RLS Policies

Feature	Priority	Test Type	Missing Scenarios	Failure Impact
Database Migrations / RLS Policies	MEDIUM	Integration	All RLS policies on 8 tables verified via SQL regex. UPDATE/DELETE policies, atomic functions (swap_active_goal, increment_rate_limit, update_streak) behavior, index effectiveness	Static SQL regex cannot verify runtime behavior. Misconfigured RLS exposes cross-tenant data.

fireAndForget.ts (16 lines)

Feature	Priority	Test Type	Missing Scenarios	Failure Impact
fireAndForget.ts	LOW	Unit	Promise rejection capture to Sentry with context string	Broken reason silently dropped. Used throughout app for non-critical writes.

4. High Priority Breakpoints (Top 10 Risks)

1. ai-checkin: 444 lines of unprotected business logic

How it breaks: The AI-powered daily check-in flow has zero tests. It handles OpenAI API calls with JSON extraction from markdown fences, task generation, database upserts, and partial failure recovery. A malformed AI response or timeout would crash the primary user workflow.

Impact: Users unable to complete daily check-in, broken task generation, stuck loading states

Severity: CRITICAL

2. activate-trial: CAS guard and INSERT fallback untested

How it breaks: The atomic CAS guard that prevents duplicate trials (`.not("status","in","(active,trialing)")`) has zero tests. If a Supabase query builder API change alters the `.not()` behavior, users could activate unlimited free trials. The INSERT fallback path for new subscriptions is also untested.

Impact: Revenue loss from infinite free trials, subscription state corruption

Severity: HIGH

3. authStore.initialize() callback never exercised

How it breaks: The `onAuthStateChange` callback is the single entry point for auth lifecycle in the app. Tests verify only the return type (`{unsubscribe}`), never the callback behavior. If `fetchProfile` fails silently in the callback, users see stale profiles indefinitely.

Impact: Stale user profiles, broken onboarding detection, incorrect entitlement display

Severity: HIGH

4. All 26 screens have zero rendering tests

How it breaks: 6,883 lines of screen code with zero component rendering tests. No `@testing-library/react-native` setup. Screen-level bugs (broken navigation, incorrect state display, accessibility violations) are invisible to the test suite.

Impact: User-facing bugs in every screen, no regression protection for UI changes

Severity: HIGH

5. Edge Function HTTP handlers never integration-tested

How it breaks: All 6 Edge Functions have their pure logic partially tested (sanitize, event mapping, etc.), but the actual `Deno.serve()` HTTP handler — including auth extraction, rate limit orchestration, and response formatting — has zero integration tests.

Impact: Auth bypass, rate limit bypass, malformed API responses

Severity: HIGH

6. Database RLS policies verified only by SQL regex

How it breaks: RLS tests use regex to confirm policies exist in migration SQL, but never verify runtime behavior. No test confirms User A cannot read User B's chat messages, sessions, or subscription data.

Impact: Cross-tenant data exposure, GDPR violation

Severity: MEDIUM

7. delete-account soft-delete + sign-out flow untested

How it breaks: The account deletion flow (soft-delete profile, then sign-out) has zero tests. If soft-delete fails midway, the account is left in a broken state — user is signed out but data is not deleted.

Impact: Inconsistent account state, compliance violation, user unable to re-register

Severity: MEDIUM

8. data-export GDPR compliance untested

How it breaks: The data export function fetches from 8+ tables with a 10,000-row cap per table. No test verifies the export payload structure, Content-Disposition headers, or handling when individual table queries fail.

Impact: Incomplete data export, GDPR non-compliance, malformed export files

Severity: MEDIUM

9. No E2E tests for critical user flows

How it breaks: Zero Detox/Maestro/Playwright configuration. Critical flows like sign-up → onboarding → check-in → timer → rating → promise have never been tested end-to-end.

Impact: Multi-step flow failures invisible to unit tests

Severity: MEDIUM

10. CI pipeline has no Deno test job

How it breaks: The 48 Deno tests for Edge Functions and RLS policies are not executed in CI. The ci.yml only runs Jest. A regression in Edge Function logic would pass CI.

Impact: Edge Function regressions deployed to production undetected

Severity: MEDIUM

5. What Was Checked (Audit Trail)

Files Analyzed:

- All 26 screen files in app/ (6,883 lines)
- All 5 store files in src/stores/ (1,056 lines)
- All 11 service files in src/services/ (922 lines)
- All 5 lib files in src/lib/ (228 lines)
- All 17 component files in src/components/ (1,498 lines)
- All 6 Edge Function files in supabase/functions/ (1,564+ lines)
- 4 database migration files in supabase/migrations/ (486 lines)
- All 33 test files (6,171 lines, 509 tests)
- jest.config.js, jest.setup.js, jest.fix-expo.js
- .github/workflows/ci.yml (104 lines)
- package.json, tsconfig.json, app.json

Test Files Mapped:

- src/stores/__tests__/ — 7 files (authStore, chatStore, goalStore, sessionStore x2, subscriptionStore x2)
- src/services/__tests__/ — 10 files (auth, chat, checkin, goals, profile, promises, sessions, shields, subscription, tasks)
- src/lib/__tests__/ — 6 files (mmkv, sentry, timeout, routeProtection, sessionRating, timerLogic)
- src/utils/__tests__/ — 3 files (constants, time, time.additional)
- src/hooks/__tests__/ — 1 file (useAppState)
- src/components/__tests__/ — 3 files (Button, Card, ErrorBoundary)
- supabase/functions/ — 3 Deno test files (ai-chat, revenuecat-webhook, rls_policy)

Assumptions Made:

- No tests were executed. All findings from static code analysis and test file mapping.
- Coverage percentages for src/ from the last recorded Jest --coverage run (74.40% stmt).
- Edge Function coverage estimated from pure-function test count vs total branch count.
- Deno tests (48 tests) are written but NOT executed in CI — marked as available but unverified.
- Branch counts are manual approximations based on if/else/switch/try-catch/ternary analysis.
- @testing-library/react-native is listed in devDependencies but no screen tests use it.

6. Final Verdict

Can this system safely go to production?

CONDITIONAL YES — with known risks accepted and mitigations in place.

Justification (direct, no hedging):

- The core business logic layer (stores + services) is well-tested at 87%+ coverage with 248 tests across 17 suites. Critical paths like subscription management, session persistence, chat rate limiting, and goal state transitions have comprehensive unit tests.
- Recent improvements have closed previously-critical gaps: REFUND webhook handler added, prompt injection defenses hardened, restoreTimerFromMMKV fully branch-tested, signUp tested, route protection logic verified, clarification loop capped, CI coverage threshold enforced at 70%.
- The remaining HIGH-risk items are: (1) ai-checkin Edge Function with 444 lines and 0 tests, (2) activate-trial CAS guard untested, (3) authStore.initialize() callback behavior untested, (4) zero screen rendering tests.
- The 48 Deno tests for Edge Functions and RLS policies exist but are NOT integrated into CI. Until a `deno test` job is added to ci.yml, these provide documentation value only.
- Zero E2E tests means multi-step user flows (onboarding → check-in → timer → rating) are only validated by the sum of their unit tests, not as integrated journeys.

Minimum gates before full production confidence:

1. Add Deno test job to CI pipeline — HIGH priority, BLOCKING
2. Write integration tests for ai-checkin (all 21 branches) — HIGH priority, BLOCKING
3. Write tests for activate-trial CAS guard + INSERT fallback — HIGH priority, BLOCKING
4. Test authStore.initialize() callback behavior (invoke the subscription callback) — HIGH priority, HIGH
5. Write tests for delete-account soft-delete flow — MEDIUM priority, HIGH
6. Write tests for data-export payload structure — MEDIUM priority, HIGH
7. Set up @testing-library/react-native and test check-in.tsx — MEDIUM priority, MEDIUM
8. Add live RLS integration tests (User A cannot read User B's data) — MEDIUM priority, MEDIUM
9. Set up E2E framework (Maestro recommended for React Native) — LOW priority, MEDIUM
10. Write fireAndForget.ts tests — LOW priority, LOW